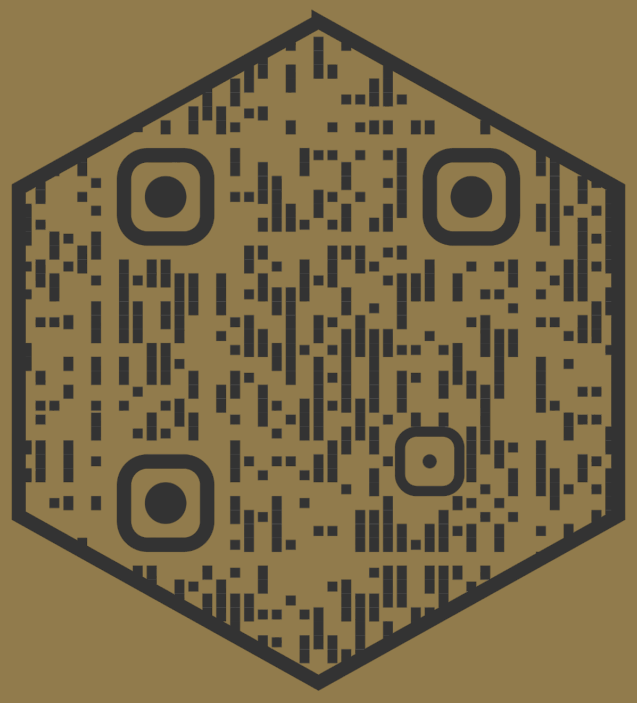




Transition of Language in the Catholic Church: Analysis of Linguistic Shifts in Papal Encyclicals Before and After Vatican II

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Abstract

This study examines linguistic shifts in papal encyclicals before and after the Second Vatican Council (1962-1965), revealing how language in papal encyclical reflects evolving philosophical perspectives within the Catholic Church. Using computational text analysis methods including word frequency analysis, topic modeling, and cosine similarity measures, we identify significant transitions from divine-centered to more human-centered terminology across different pontificates. Our findings demonstrate that the encyclicals of post-Vatican II popes show increased emphasis on social justice, human dignity, and community-oriented themes. The research provides quantitative evidence of how language in official Church documents mirrors broader philosophical developments in contemporary Catholicism.

Research Questions: Language Reflects Philosophy

- How has the frequency of human-centered vs. divine-centered terminology changed over time in papal documents?
- What shifts in thematic relationships between key concepts can be observed across different pontificates?
- How has the Second Vatican Council influenced these linguistic changes?

Methods

Word Frequency Analysis: We calculated the relative frequency of key terms across papal encyclicals comparing pre- and post-Vatican II popes.

Topic Modeling: Latent Dirichlet Allocation was used to identify dominant topics in each pontificate, revealing shifting thematic priorities over time.

Cosine Similarity Analysis: We measured the semantic proximity between conceptually opposed word clusters to track changing conceptual relationships.

Word Frequency

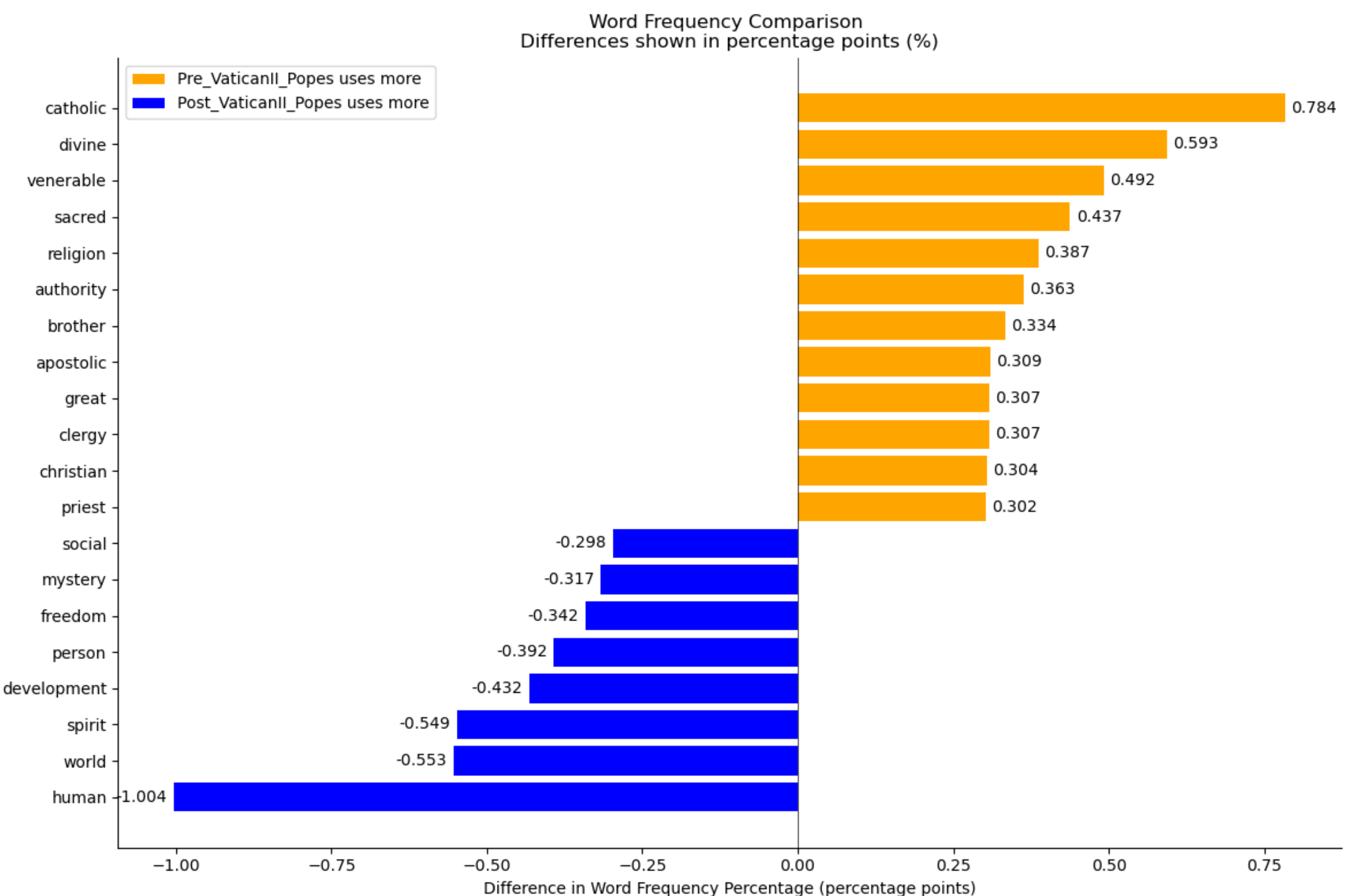


Figure 1. Comparative word frequency analysis of human-centered terminology versus divine-centered terminology can be observed in post-Vatican II papal encyclicals, demonstrating increased human-centered language.

Topic Models

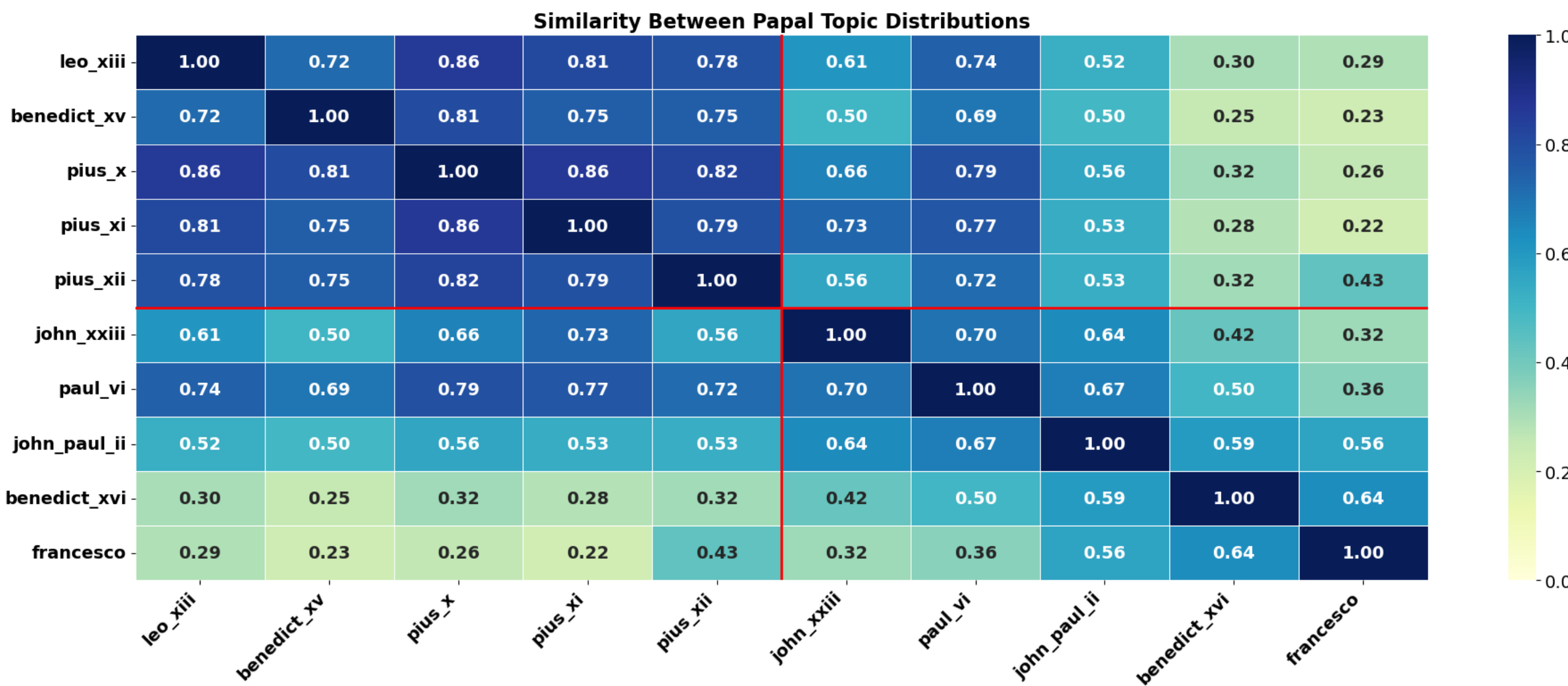


Figure 2. This similarity plot quantifies the cosine similarity between topic distributions across pontificates. It reveals strong thematic cohesion among pre-Vatican II popes (upper left quadrant) and increasing divergence in later pontificates (bottom right quadrant), suggesting a gradual rather than abrupt transformation in theological discourse following Vatican II.

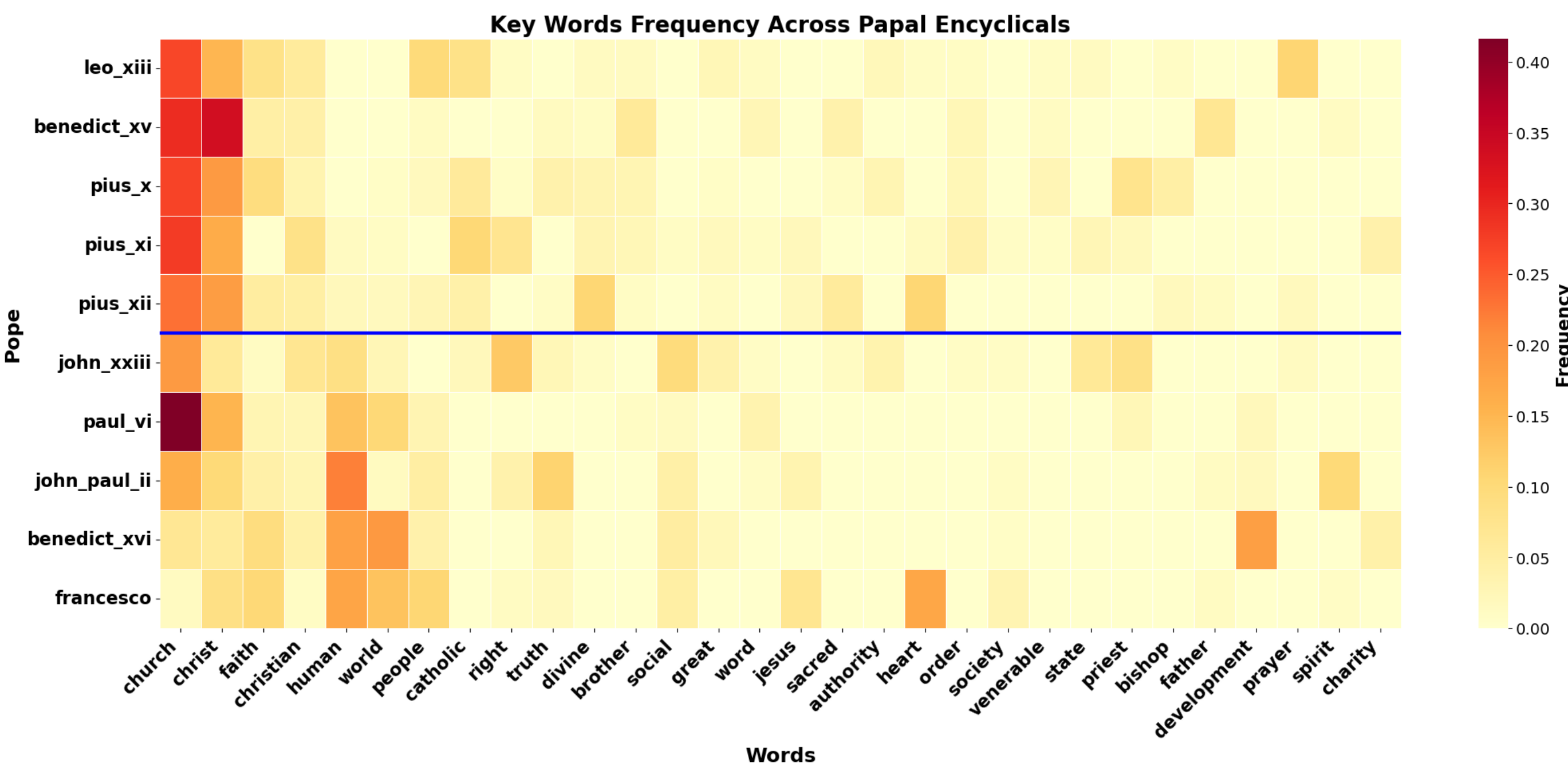


Figure 3. This heatmap visualizes the relative frequency of encyclical keywords across pontificates, revealing distinct vocabulary patterns before and after the Council. Pre-Vatican II encyclicals show greater emphasis on concepts like "church" and "doctrine," while post-Vatican II texts demonstrate increased usage of terms like "human," "social," and "development." This reflects a shift from ecclesiastical to humanistic theological framing. One can also see the favorite words across popes.

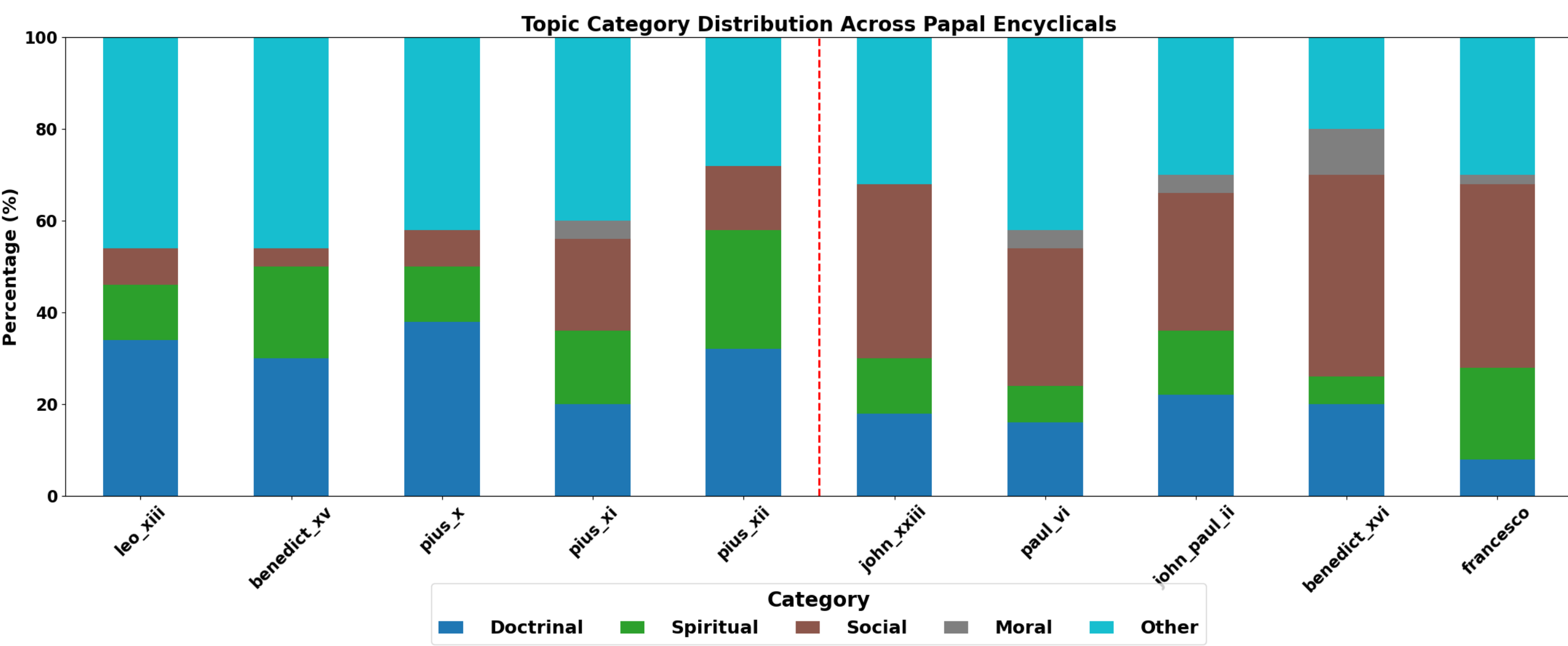


Figure 4. This stacked bar chart displays the proportional distribution of five topic categories (Doctrinal, Spiritual, Social, Moral, and Other) across pontificates calculated with topic model scores across popes. We observe a notable decrease in doctrinal content and corresponding increases in social and moral topics after Vatican II, quantitatively supporting historical interpretations of the Second Vatican Council's pastoral reorientation.

Cosine Similarity

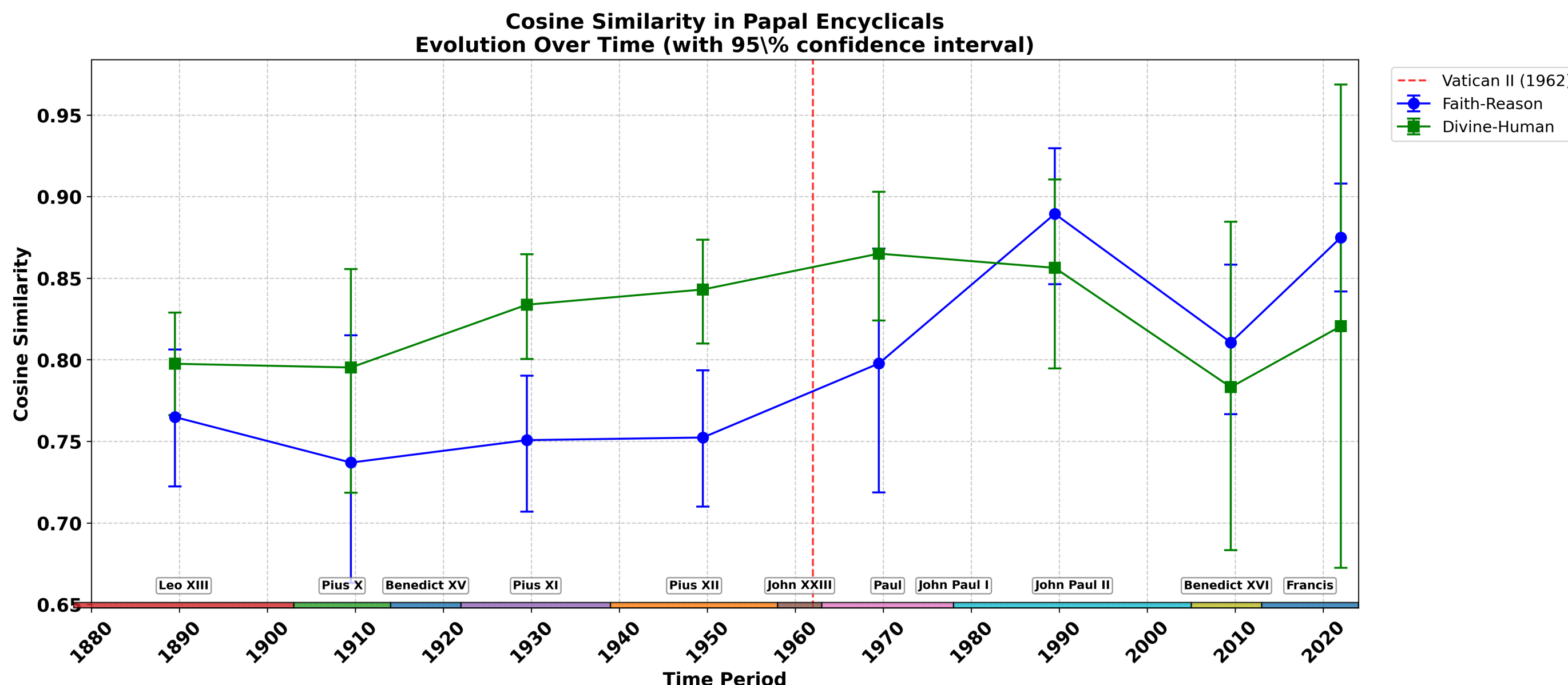


Figure 5. This figure displays the changing cosine similarity between Faith-Reason concepts and Divine-Human concepts. Faith-Reason cosine similarity shows a notable increase after Vatican II with Divine-Human cosine similarity steadily increasing, though before Vatican II. The plot is presented with bootstrapped 95% confidence intervals.

Representative Quotes: Before Vatican II

Faith-Reason Relationship
Aeterni Patris (1879)

"Faith frees and saves reason from error, and endows it with manifold knowledge."

— Pope Leo XIII

Divine-Human Relationship
Immortale Dei (1885)

"We are bound absolutely to worship God in that way which He has shown to be His will."

— Pope Leo XIII

Representative Quotes: After Vatican II

Faith-Reason Integration
Fides et Ratio (1998)

"Faith and reason are like two wings on which the human spirit rises to the contemplation of truth; and God has placed in the human heart a desire to know the truth."

— Pope Saint John Paul II

Divine-Human Relationship
Deus Caritas Est (2005)

"In the living community of believers, we experience the love of God. We perceive his presence and we thus learn to recognize that presence in our daily lives."

— Pope Benedixt XVI

Conclusion

My analysis reveals **significant linguistic shifts** in papal encyclicals, demonstrating a measurable increase in human-centered terminology following Vatican II while maintaining fundamental doctrinal continuity. The **validity** of these findings is strengthened by the triangulation of three textual analysis methods that converge on similar conclusions, while representative quotes further corroborate these results. A important **limitation** of my study is that I only considered the texts of papal encyclicals. Future research may incorporate homily texts to provide a more comprehensive analysis of thematic relationships. This methodological study offers a **valuable complement** to traditional hermeneutical analyses by quantifying subtle but significant shifts in religious themes that reflect broader theological and philosophical transitions within the Catholic Church.